WORKED SOLUTION

STEP I 2013, Question 7

Pure • First-Order Differential Equations

STEP I 2013, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q7

The key idea

In (i) and (ii) the right-hand side depends only on the ratio y/x, so writing $y = ux$ turns the equation into one in which the variables separate. Part (iii) breaks that pattern, because x^2/y is not a function of y/x alone. The cure is to ask what power of x the solution "should" behave like: balancing the two terms suggests $y = u\,x^{3/2}$, and then everything separates again.

Solution

Part (i)

Put $y = ux$, so that by the product rule $\dfrac{\mathrm{d}y}{\mathrm{d}x} = u + x\dfrac{\mathrm{d}u}{\mathrm{d}x}$. Since $x/y = 1/u$ and $y/x = u$, the equation becomes

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + u \quad \Longrightarrow \quad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u}.$$

The u terms cancel, which is what makes the substitution work. Separating the variables (we may divide by x since $x > 0$),

$$\int u\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x \quad \Longrightarrow \quad \tfrac{1}{2}u^2 = \ln x + C,$$

with $\ln x$ rather than $\ln|x|$ because $x > 0$. When $x = 1$, $y = 2$, so $u = y/x = 2$, giving $2 = 0 + C$. Hence

$$u^2 = 4 + 2\ln x.$$

Now $y > 0$ and $x > 0$ force $u > 0$, so we take the positive root, $u = \sqrt{4 + 2\ln x}$, and

$$y = x\sqrt{4 + 2\ln x}.$$

This needs $4 + 2\ln x > 0$ (we need $y > 0$, not merely $y \geq 0$), that is $\ln x > -2$, or $x > \mathrm{e}^{-2}$. This is the required result.

Part (ii)

The right-hand side is again a function of y/x, so we use the same substitution $y = ux$:

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + 2u \quad \Longrightarrow \quad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + u = \frac{1+u^2}{u}.$$

This time the u terms do not cancel, but the variables still separate:

$$\int \frac{u}{1+u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x \quad \Longrightarrow \quad \tfrac{1}{2}\ln(1+u^2) = \ln x + C,$$

where the left-hand integral is of the form $\frac{1}{2}\int f'/f$ and $1+u^2 > 0$ needs no modulus. At $x = 1$ we have $u = 2$, so $C = \frac{1}{2}\ln 5$. Multiplying by 2 and exponentiating,

$$1 + u^2 = 5x^2, \qquad \text{so} \qquad u = \sqrt{5x^2 - 1}$$

(positive root, as $u > 0$). Therefore

$$y = x\sqrt{5x^2 - 1} \qquad (x > \tfrac{1}{\sqrt{5}}),$$

the restriction coming from $5x^2 - 1 > 0$ with $x > 0$.

Part (iii)

Now $y = ux$ gives $x^2/y = x/u$, which still contains x on its own, so the equation does not separate. We want a substitution under which both terms on the right scale the same way.

> **?** **Ask yourself** If y behaves like a power x^k, which value of k makes $\dfrac{x^2}{y}$ and $\dfrac{2y}{x}$ the same size?

We need x^{2-k} and x^{k-1} to match, so $2 - k = k - 1$, giving $k = \frac{3}{2}$. (The left-hand side $\mathrm{d}y/\mathrm{d}x$ then also behaves like $x^{1/2}$, so all three terms agree.) This suggests the substitution $y = u\,x^{3/2}$, for which

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x^{3/2}\frac{\mathrm{d}u}{\mathrm{d}x} + \tfrac{3}{2}u\,x^{1/2}, \qquad \frac{x^2}{y} = \frac{x^{1/2}}{u}, \qquad \frac{2y}{x} = 2u\,x^{1/2}.$$

Dividing the whole equation by $x^{1/2}$ (which is positive),

$$x\frac{\mathrm{d}u}{\mathrm{d}x} + \tfrac{3}{2}u = \frac{1}{u} + 2u \quad \Longrightarrow \quad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + \frac{u}{2} = \frac{2+u^2}{2u}.$$

Separating,

$$\int \frac{2u}{2+u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x \quad \Longrightarrow \quad \ln(2+u^2) = \ln x + C.$$

At $x = 1$, $u = y/x^{3/2} = 2$, so $C = \ln 6$, and hence $2 + u^2 = 6x$. Taking the positive root, $u = \sqrt{6x-2}$, so

$$y = x^{3/2}\sqrt{6x-2} = \sqrt{6x^4 - 2x^3} \qquad (x > \tfrac{1}{3}),$$

where $x > \frac{1}{3}$ is needed for $6x - 2 > 0$.

An alternative for (iii). Multiplying the equation by $2y$ gives, with $z = y^2$,

$$\frac{\mathrm{d}z}{\mathrm{d}x} - \frac{4}{x}z = 2x^2,$$

a linear equation with integrating factor x^{-4}. Then $(x^{-4}z)' = 2x^{-2}$, so $x^{-4}z = -2/x + 6$ using $z(1) = 4$, giving $y^2 = 6x^4 - 2x^3$ as before. The same trick ($z = y^2$) also solves (i) and (ii).

Checks

- Each answer gives $y = 2$ at $x = 1$: $1 \cdot \sqrt{4}$, $1 \cdot \sqrt{4}$ and $\sqrt{6-2}$.
- Differentiate the answer to (ii) and substitute back. From $y^2 = 5x^4 - x^2$ we get $2yy' = 20x^3 - 2x$, so $y' = (10x^3 - x)/y$. Meanwhile $\dfrac{x}{y} + \dfrac{2y}{x} = \dfrac{x^2 + 2y^2}{xy} = \dfrac{10x^4 - x^2}{xy}$, which is the same.
- At $x = 2$ the three solutions give $2\sqrt{4 + 2\ln 2} \approx 4.642$, $2\sqrt{19} \approx 8.718$ and $4\sqrt{5} \approx 8.944$. As expected, the larger coefficient of y/x in (ii) makes y grow faster than in (i).

Common slips

- Forgetting the u in $\dfrac{\mathrm{d}y}{\mathrm{d}x} = u + x\dfrac{\mathrm{d}u}{\mathrm{d}x}$, so that the equation in u comes out wrong.
- Applying the initial condition to y instead of u: at $x = 1$ the two agree here, but in (iii) you need $u = y/x^{3/2}$ in general.
- Leaving out the domain. The square roots must be real and positive, which gives $x > \mathrm{e}^{-2}$, $x > 1/\sqrt{5}$ and $x > \frac{1}{3}$.

Beyond STEP

Equations whose right-hand side depends only on y/x are called *homogeneous*, and the substitution $y = ux$ always separates them. Part (iii) is a scaling symmetry in disguise: the equation is unchanged under $x \mapsto \lambda x$, $y \mapsto \lambda^{3/2} y$, and such a scaling symmetry always suggests a substitution that makes the equation separable. Equations of the form $y' + P(x)y = Q(x)y^n$, which become linear under $z = y^{1-n}$, are called Bernoulli equations; all three parts here are of this type with $n = -1$. Boyce and DiPrima, *Elementary Differential Equations*, covers both families in its chapter on first-order equations.

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